Aayush Iyengar – Project Portfolio

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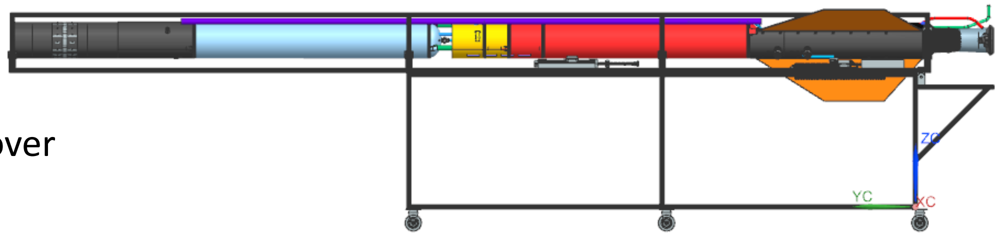
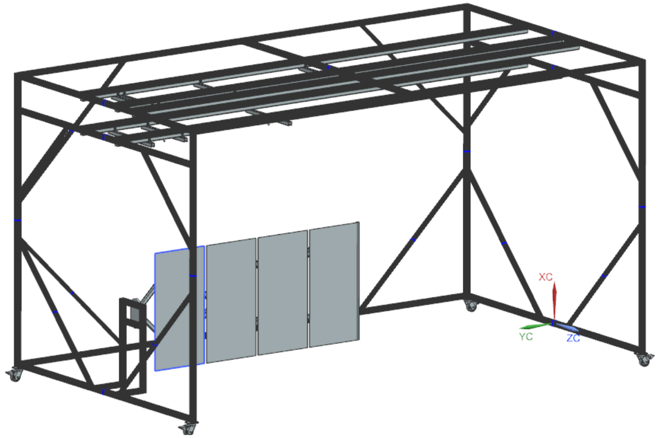
About Me

- Senior at Purdue University pursuing a B.S in Aeronautical/Astronautical Engineering
  - Minor in AI/ML
- Previous Experience:
  - **SpaceX** - Stage 0 Propellent Systems Engineering Intern: 05/25 - 08/25
  - **Rocket Lab** - Spacecraft Mechanical Engineering Intern: 01/25 – 04/25
  - **Purdue** – Undergraduate Research Drone Docking Mechanism Lead: 01/24 – 12/24
  - **SpaceX** – Launch Engineering Intern: 05/24 – 08/24
- Currently the Breakover Cart **RE** @ PSP
- Extensive experience with tools covering design (**NX** and **SolidWorks**), analysis (**ANSYS** and **CAESAR II**), and modeling (**Python** and **MATLAB**/**Simulink**).
- Hobbies: Basketball, Jazz, Entrepreneurship, Poker

My Project Background

- Purdue Space Program:
  - Breakover Cart – Owned the **design** & **analysis** of the strongback + breakover mechanism. Project is currently gearing up for CDR.
- SpaceX (2025)
  - Offload Station Redesign – Completed **design** & **analysis** through CDR.
  - Pad 2 Flame Diverter Shielding – Completed **design**, **analysis**, and **installation.**
  - Masseys Waterpad Rebuild – Undergoing activation after installation of all major components.
- Rocket Lab
  - Solar Array Offloader – **Conceptualized**, **built** and **tested** dynamic gravity offloading system.
- SpaceX (2024)
  - GCH4 Pump supply and venting system - **Redesigned** and **installed** modifications.
  - Pad 1 Flame Diverter Blast Protection – **Installed**
- Undergraduate Research
  - Drone Docking and Mechanism – **Design**, **installation**, and **testing**.





In-Depth Project Breakdown

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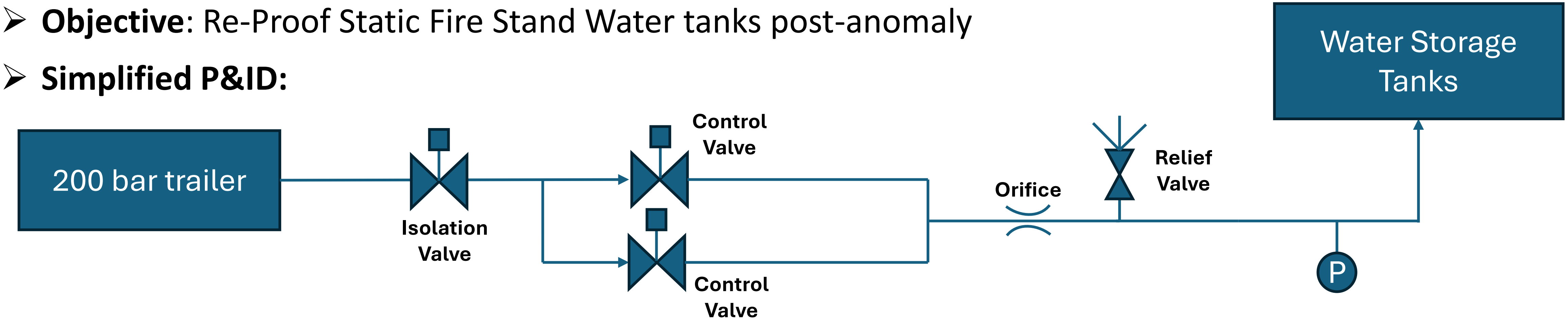
SpaceX Water Tank Proof Blowdown Test

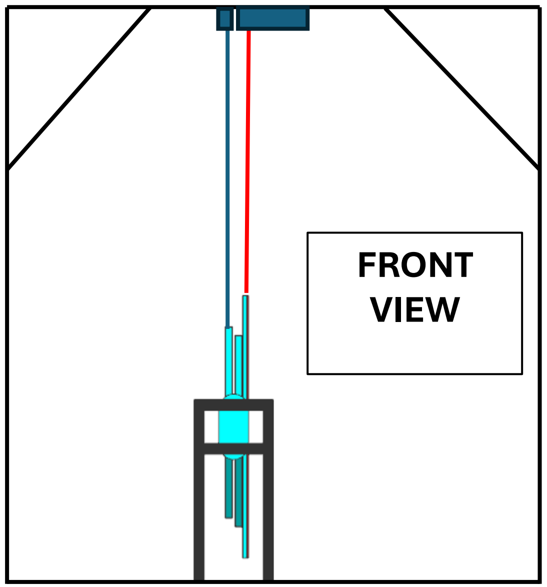
- **Objective**: Re-Proof Static Fire Stand Water tanks post-anomaly
- **Simplified P&ID:**



- Determined **press path,** sized **press time** (function of mass flow rate set by orifice), **trailer volume required**, and **site clear radius** (function of tank ullage %).

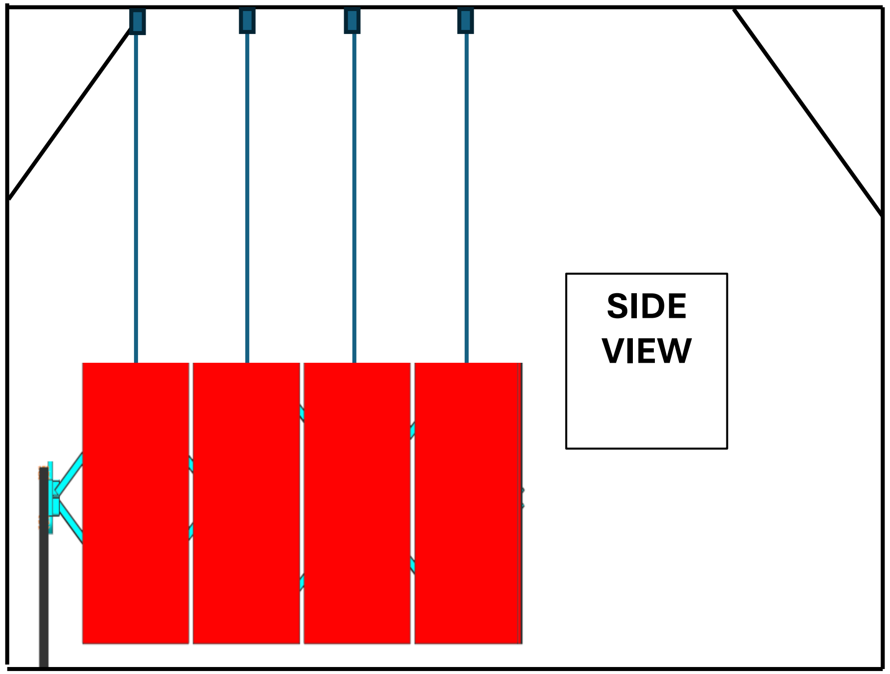
Rocket Lab - Solar Array Offloader

Objective: Design, manufacture, and activate a mechanism to test the deployment characteristics of a new solar array deployment architecture for Rocket Lab.

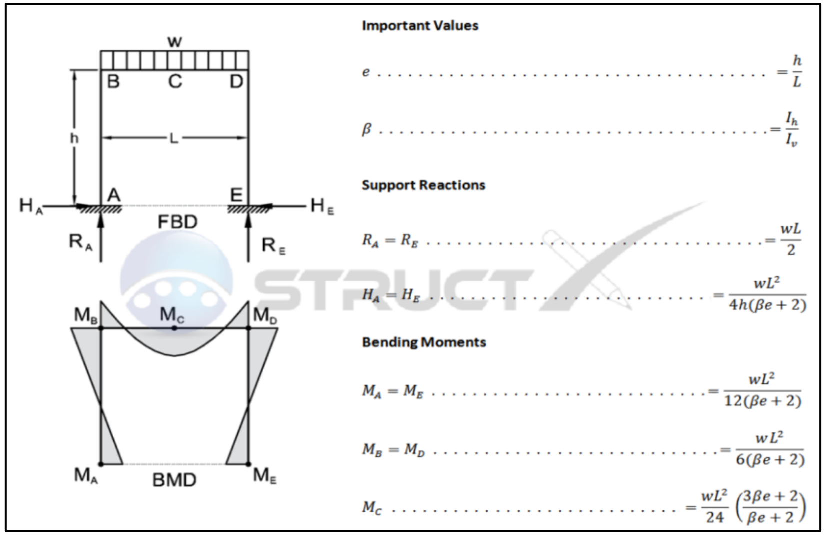


Notable Achievements:

- Drove design from requirements gathering to conceptual design to installation to testing.
- Tested and proved the functionality of linear deployment mechanism for Rocket Lab.
- Created linear deployment mechanism and solar array gravity offloader sizing calculators for future use and implementation.
- Manufactured the offloader and all sub-components and performed deployment testing of the linear deployment mechanism.



Solar Array Offloader - Analysis



- Performed Beam sizing/analysis via StructX Frame Hand Calcs
  - Offloader has a uniformly distributed load (weight), and individual point loads (each offloading point) so the true loading is a mixture of the two boundary conditions
- Buckling Analysis of long vertical members:
- Need for Stiffeners – driven by operational requirement to support applied lateral forces to different sections of the offloader
- Tipover Hand Calc
  - System COM relative to a tipping point where system total moments are calculated into the System's static tipping point
  - Allows us to back out the additional force that is needed to cause a tip.
    - $F_{tip} = \frac{(m_{Tot} * g) * r_{COM}}{h}$

Inputs	Quantity	Value	Unit	Notes
	Length	216	in	
	Height	120	in	
	Dist. Load	250	lbf	
	Dist. Load/L	1.157407	lbf/in	
	I_h	0.551727	in^4	
	I_v	0.551727	in^4	
	Area Vert.	0.9375	in^2	
	Area Horiz.	0.9375	in^2	
	E (al 6063)	10000000	PSI	
	Failure Stress	13333.33	PSI	yield stress/SF
	y (for bending stress)	1	in	
	K_col	0.5		fixed-fixed

Calculations	Quantity	Value	Unit	Notes
	Beta	1	n/a	
	e	1.8	n/a	
	R_A	125	lbf	
	R_E	125	lbf	
	H_A	29.60526	lbf	
	H_E	29.60526	lbf	
	M_A	1184.211	lbf-in	
	M_E	1184.211	lbf-in	
	M_B	2368.421	lbf-in	
	M_D	2368.421	lbf-in	
	M_C	4381.579	lbf-in	

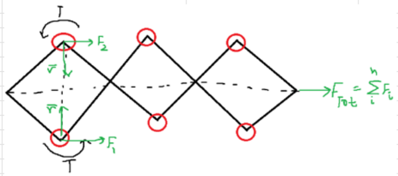
Outputs	Quantity	Value	Unit	Notes
	Max Bending Stress	7941.571	psi	
	Max Compressive Stress	133.3333	psi	
	MOS	0.678929		
	Critical Buckling Stress	16134.3	psi	

Deployment Mech. Tension Sizing

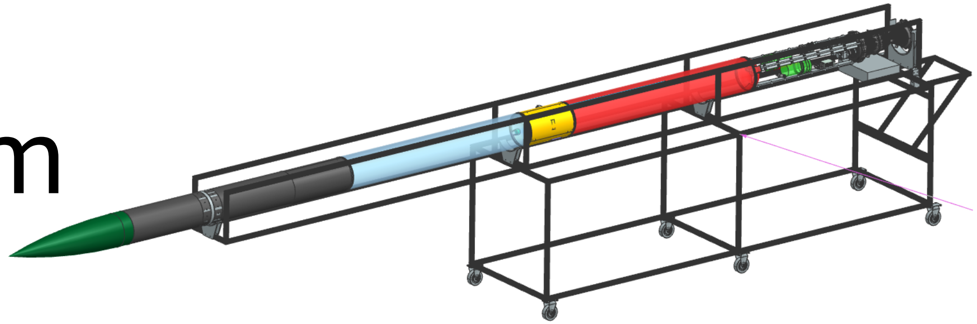
Inputs	Quantity	Value	Unit	Notes
	# Springs	6	n/a	
	K_spring	0.2275	Nm/rad	*can iteratively change
	theta_deployed	135	deg	
	linkage length	4	ft	

Calculations	Quantity	Value	Unit	Notes
	theta deployed	2.356	rad	
	radial arm	2.000	ft	*half of linkage length
	radial arm	0.610	m	
	Moment per linkage	0.536	Nm	
	Force Per linkage	1.758	N	

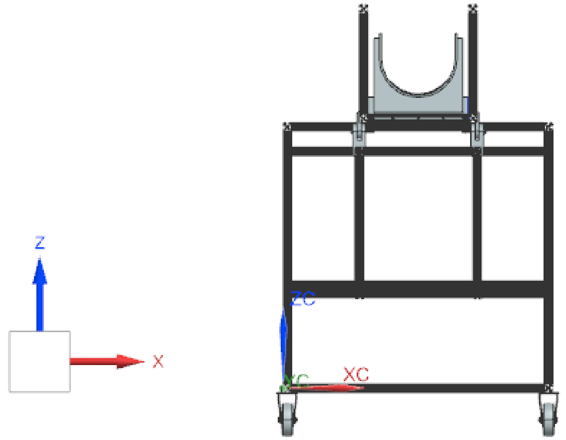
Outputs	Quantity	Value	Unit	Notes
	Moment from all linkages	3.216	Nm	
	Force from all linkages	10.551	N	



PSP - Breakover Cart Mechanism

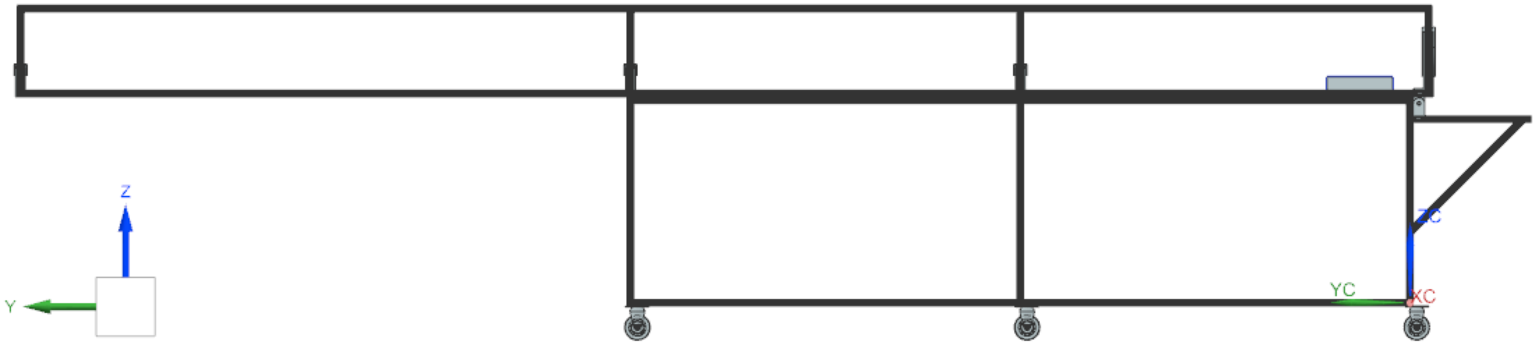


Objective: PSP’s next rocket is ~22ft tall, making it too tall to lift/work on while vertical. A system is needed to allow RE’s to work on the rocket prior to going vertical and safely transition the rocket if need be.

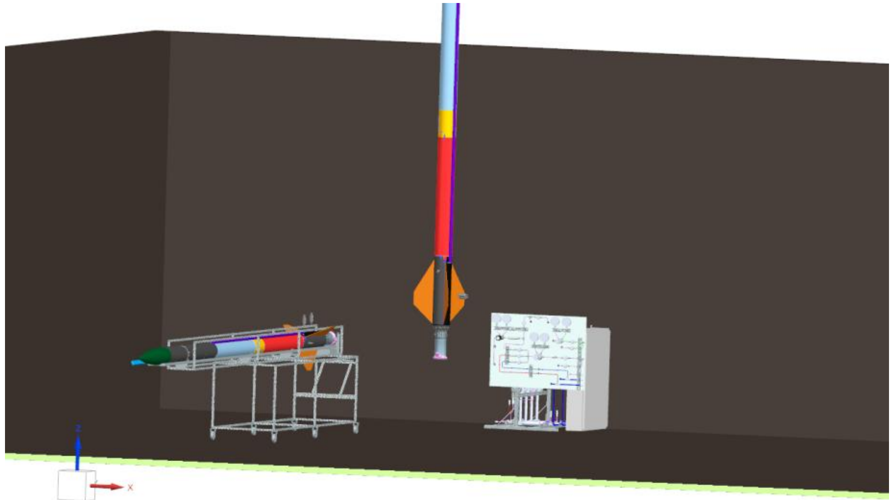


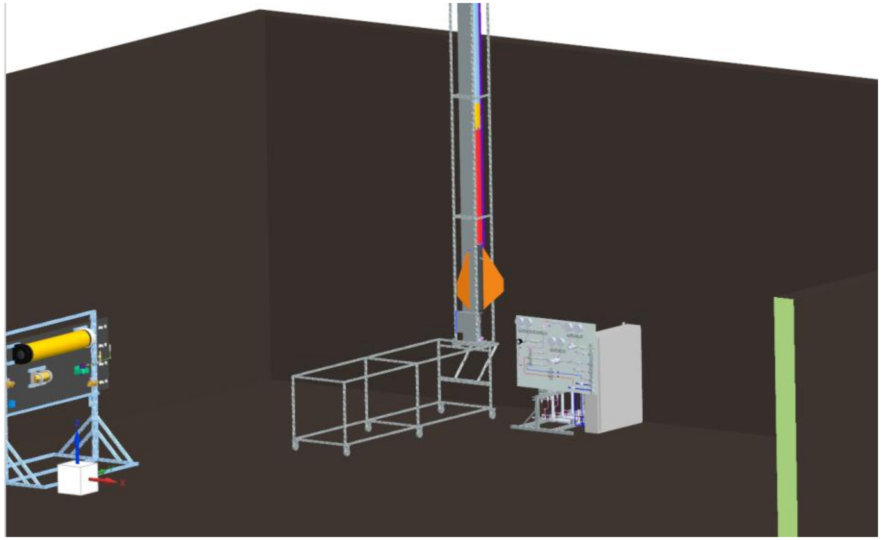
Notable Achievements:

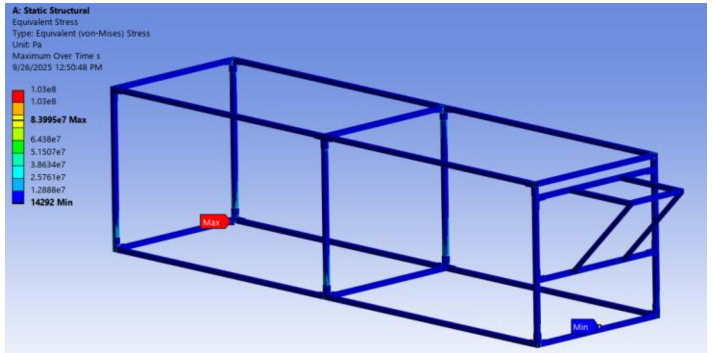
- Drove design from requirements gathering to conceptual design to build.
- Owned the top level design decisions and led a team of 5 students to design and analyze all components of the system.
- Designed and analyzed all rocket-cart interface components before getting into manufacturing.



Breakover Cart Mechanism - Analysis

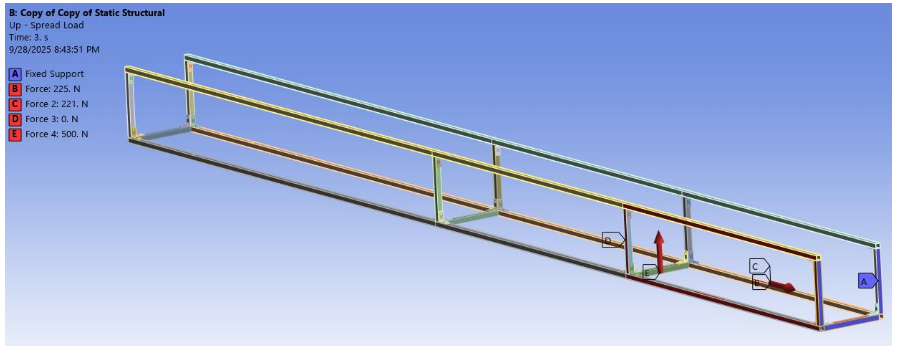






Step	Load Applied
1	Fg Strongback + Rocket, applied at 6X cart representative locations.
2	Fg + 500N Length Direction Push, applied at all faces.
3	Fg + 500N Width Direction Push, applied at all faces
4	Fg + 500N Width + Length Direction Push, applied at all faces.

Results Summary	
Max Stress	8.4E7 Pa
Yield Material Allowable W/ SF = 2	1.38E8 Pa
Ult Material Allowable W/ SF = 3	1.03E8 Pa
Yield MoS	**+ 0.64**
Ult MoS	**+ 0.23**



Results Summary	
Max Stress – 5X Attachments	2.62E8 Pa
Max Stress Used – 5X Attachments	9.41E7 Pa
Max Stress – 2X Attachments	2.64E8 Pa
Max Stress Used – 2X Attachments	9.35E7 Pa
Yield Material Allowable W/ SF = 2	1.38E8 Pa
Ult Material Allowable W/ SF = 3	1.03E8 Pa
Driving Yield MoS	**+ 0.5**
Driving Ult MoS	**+ 0.1**

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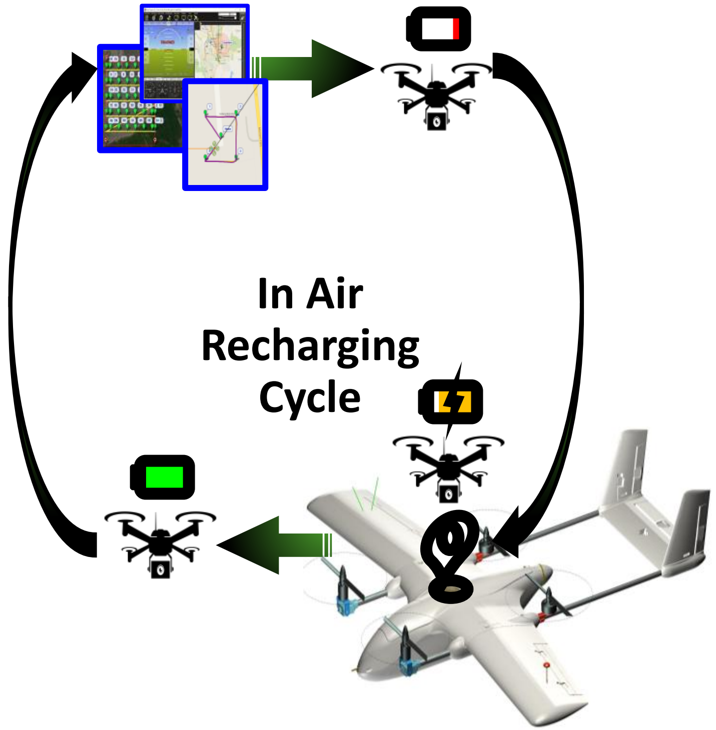
Autonomous Drone In-Air Charging

- Automation and Optimization Lab

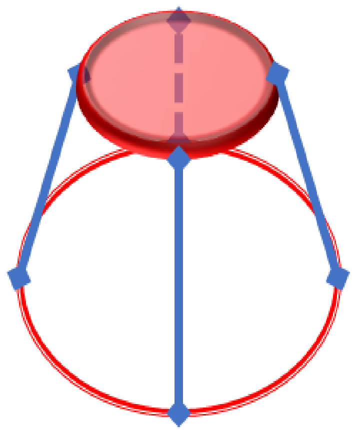


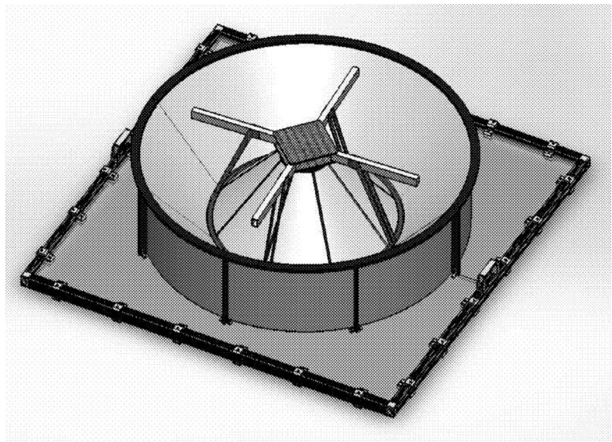
Objective: Developing an aerial recharging platform for a specific drone combination to ensure uninterrupted, autonomous flight during the drone's cycle. This provides benefits for extended flight durations, continuous mission capabilities, and environmental benefits.

Notable Achievements:

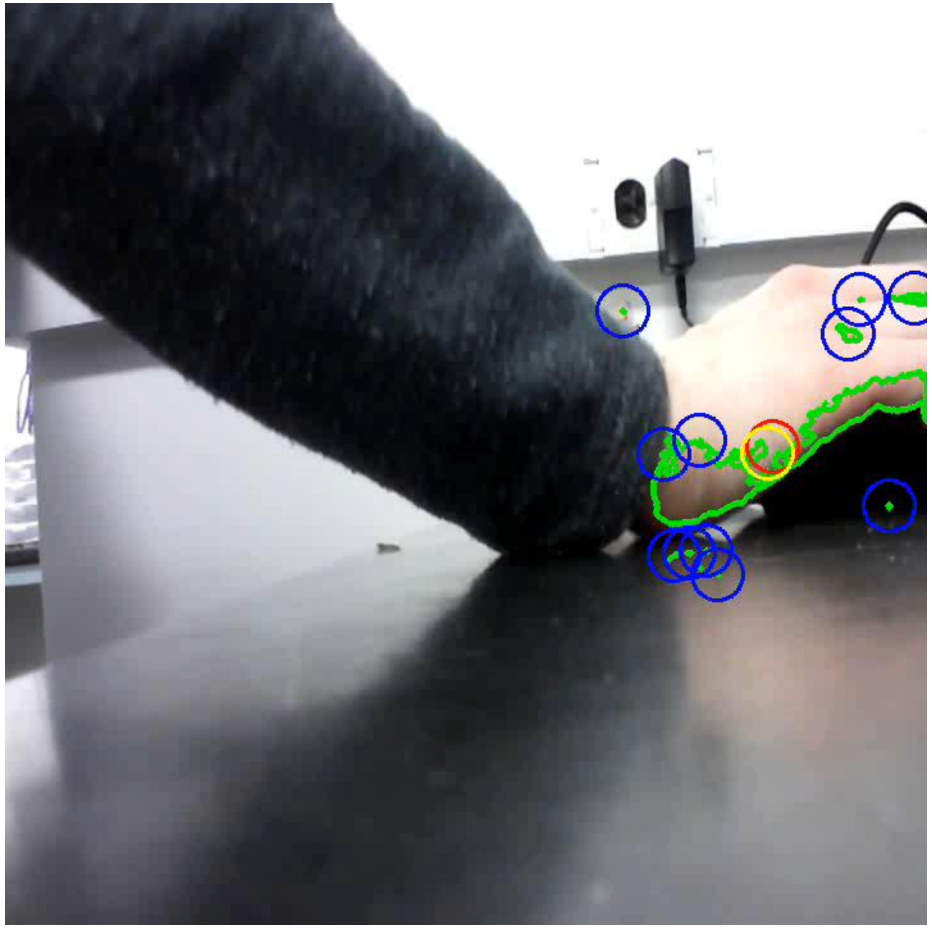
- Design, analysis and integration of drone docking mechanism.
- Development and implementation of vision-based target detection algorithm.
- Completed a test of 2 charge cycles for the small drone on the larger drone.
- Published results in AIAA article.

Landing 3D printed funnel & the landing funnel CAD Design

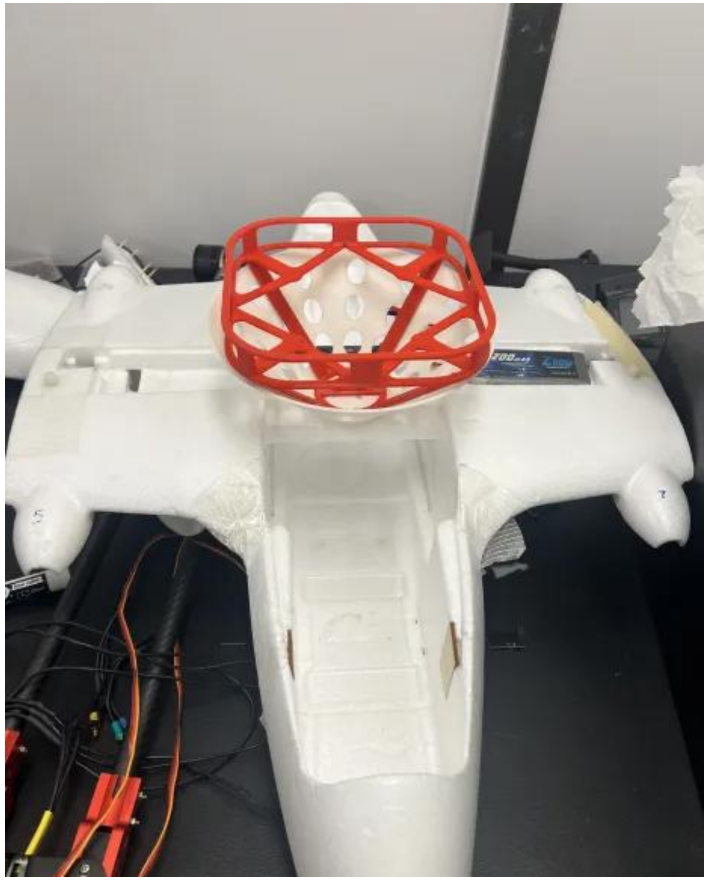




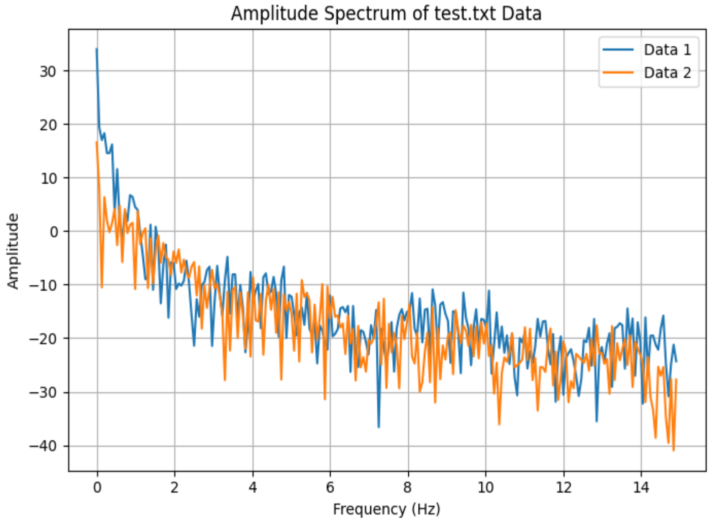
Autonomous Drone In-Air Charging - Analysis



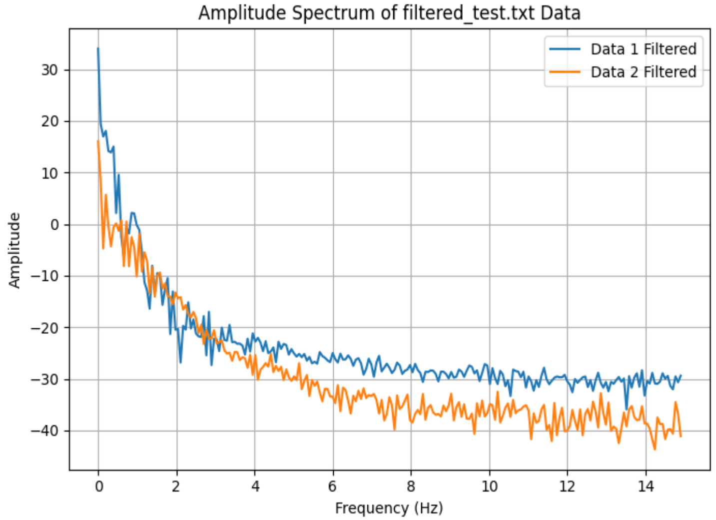
Centroid Detection Example



Integrated Charging and Landing System



Unfiltered Data



Filtered Data

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